

Speed, Distance & Time

Answer Key & Detailed Solutions — Questions 22 to 41

Question 22

A motorbike leaves point A at 1 pm and moves towards point B at a uniform speed. A car leaves point B at 2 pm and moves towards point A at a uniform speed which is double that of the motorbike. They meet at 3:40 pm at a point which is 168 km away from A. What is the distance, in km, between A and B?

A) 364

✓ B) 378 ← Correct Answer

C) 380

D) 388

Solution:

Speed of motorbike = s km/hr; Speed of car = $2s$ km/hr.

Time of Travel:

- Motorbike starts at 1 pm, car starts at 2 pm, they meet at 3:40 pm.
- Time traveled by motorbike = 2 hrs 40 min = $\frac{8}{3}$ hours.
- Time traveled by car = 1 hr 40 min = $\frac{5}{3}$ hours.

Distance Traveled by Motorbike = 168 km (given).

$$\rightarrow 168 = s \times \left(\frac{8}{3}\right) \rightarrow s = 168 \times \frac{3}{8} = 63 \text{ km/hr.}$$

Speed of Car = $2 \times 63 = 126$ km/hr.

Distance Traveled by Car = $126 \times \left(\frac{5}{3}\right) = 210$ km.

Total Distance A to B = $168 + 210 = 378$ km.

Question 23

In a car race, car A beats car B by 45 km, car B beats car C by 50 km, and car A beats car C by 90 km. The distance (in km) over which the race has been conducted is

A) 475

✓ B) 450 ← Correct Answer

C) 500

D) 550

Solution:

Let race distance = D km.

Speed ratios from race results:

- $S_A / S_B = D / (D - 45) \dots(1)$
- $S_B / S_C = D / (D - 50) \dots(2)$
- $S_A / S_C = D / (D - 90) \dots(3)$

Multiply (1) $\times$ (2) = (3):

$$[D/(D-45)] \times [D/(D-50)] = D/(D-90)$$

$$D/(D-45) \times D/(D-50) = D/(D-90)$$

$$D^2 / [(D-45)(D-50)] = D / (D-90)$$

$$D(D-90) = (D-45)(D-50)$$

$$D^2 - 90D = D^2 - 95D + 2250$$

$$5D = 2250$$

$$D = 450 \text{ km.}$$

Question 24

A and B are two railway stations 90 km apart. A train leaves A at 9:00 am, heading towards B at a speed of 40 km/hr. Another train leaves B at 10:30 am, heading towards A at a speed of 20 km/hr. The trains meet each other at

- A) 11:45 AM
- B) 11:20 AM
- ✓ C) 11:00 AM ← Correct Answer
- D) 10:45 AM

Solution:

Train A starts at 9:00 am; Train B starts at 10:30 am.

Distance covered by Train A before Train B starts (1.5 hrs):

$$= 40 \times 1.5 = 60 \text{ km.}$$

Remaining distance at 10:30 am = $90 - 60 = 30 \text{ km.}$

Relative speed (trains moving towards each other) = $40 + 20 = 60 \text{ km/hr.}$

Time to meet = $30 / 60 = 0.5 \text{ hours} = 30 \text{ minutes.}$

Meeting time = 10:30 am + 30 min = 11:00 AM.

Question 25

Two places A and B are 45 kms apart and connected by a straight road. Anil goes from A to B while Sunil goes from B to A. Starting at the same time, they cross each other in exactly 1 hour 30 minutes. If Anil reaches B exactly 1 hour 15 minutes after Sunil reaches A, the speed of Anil, in km per hour, is

- 1) 18
- 2) 16
- 3) 14
- ✓ 4) 12 ← Correct Answer

Solution:

They meet after $t_{\text{meet}} = 1.5 \text{ hours.}$

Using the formula: $t_{\text{meet}}^2 = t_{\text{A_rem}} \times t_{\text{S_rem}}$

and $t_{\text{A_rem}} = t_{\text{S_rem}} + 1.25$ (Anil takes 1 hr 15 min longer after meeting).

$$1.5^2 = (t_{\text{S_rem}} + 1.25) \times t_{\text{S_rem}}$$

$$2.25 = t_{\text{S_rem}}^2 + 1.25 \cdot t_{\text{S_rem}}$$

$$4t^2 + 5t - 9 = 0 \text{ (multiply by 4)}$$

$$(t - 1)(4t + 9) = 0 \rightarrow t_{\text{S_rem}} = 1 \text{ hour.}$$

$$t_{\text{A_rem}} = 1 + 1.25 = 2.25 \text{ hours.}$$

$$\text{Total time for Anil} = 1.5 + 2.25 = 3.75 \text{ hours.}$$

$$\text{Anil's speed} = 45 / 3.75 = 12 \text{ km/hr.}$$

Question 26

A and B are two points on a straight line. Ram runs from A to B while Rahim runs from B to A. After crossing each other, Ram and Rahim reach their destination in one minute and four minutes, respectively. If they start at the same time, then the ratio of Ram's speed to Rahim's speed is

A) $1/2$

B) $\sqrt{2}$

✓ C) 2 ← Correct Answer

D) $2\sqrt{2}$

Solution:

Using the standard formula: when two runners start simultaneously from A and B, move toward each other, meet, then reach their respective destinations in times t_1 and t_2 :

$$\text{Speed ratio} = \sqrt{(t_2 / t_1)}$$

Here t_1 (Ram's remaining time) = 1 min, t_2 (Rahim's remaining time) = 4 min.

$$s_{\text{Ram}} / s_{\text{Rahim}} = \sqrt{(4 / 1)} = \sqrt{4} = 2.$$

Question 27

Two circular tracks T1 and T2 of radii 100 m and 20 m, respectively touch at a point A. Starting from A at the same time, Ram and Rahim are walking on track T1 and track T2 at speeds 15 km/hr and 5 km/hr respectively. The number of full rounds that Ram will make before he meets Rahim again for the first time is

A) 5

✓ B) 3 ← Correct Answer

C) 2

D) 4

Solution:

They can only meet again at point A (the single touch point).

Time for Ram to complete one lap (T1, radius 100 m = 0.1 km):

$$T_{\text{Ram}} = (2\pi \times 0.1) / 15 = \pi/75 \text{ hr.}$$

Time for Rahim to complete one lap (T2, radius 20 m = 0.02 km):

$$T_{\text{Rahim}} = (2\pi \times 0.02) / 5 = 0.04\pi/5 = \pi/125 \text{ hr.}$$

$$\text{First meeting time} = \text{LCM}(\pi/75, \pi/125) = \pi / \text{GCF}(75, 125) = \pi/25 \text{ hr.}$$

$$\text{Rounds by Ram} = (\pi/25) / (\pi/75) = 75/25 = 3 \text{ rounds.}$$

Question 28

Anil, Sunil, and Ravi run along a circular path of length 3 km, starting from the same point at the same time, and going in the clockwise direction. If they run at speeds of 15 km/hr, 10 km/hr, and 8 km/hr, respectively, how much distance in km will Ravi have run when Anil and Sunil meet again for the first time at the starting point?

✓ A) 4.8 ← Correct Answer

B) 4.6

- C) 5.2
D) 4.2

Solution:

Lap times:

- $T_{\text{Anil}} = 3/15 = 1/5$ hr
- $T_{\text{Sunil}} = 3/10$ hr

Time for Anil and Sunil to meet at starting point = $\text{LCM}(1/5, 3/10)$.

$\text{LCM of fractions} = \text{LCM}(\text{numerators})/\text{GCF}(\text{denominators}) = \text{LCM}(1,3)/\text{GCF}(5,10) = 3/5$ hr.

Distance covered by Ravi in $3/5$ hr:

$$= 8 \times (3/5) = 24/5 = 4.8 \text{ km.}$$

Question 29

Two cars travel the same distance starting at 10:00 am and 11:00 am, respectively, on the same day. They reach their common destination at the same point of time. If the first car travelled for at least 6 hours, then the highest possible value of the percentage by which the speed of the second car could exceed that of the first car is

✓ A) 20 ← Correct Answer

- B) 30
C) 25
D) 10

Solution:

Car 1 travels for t_1 hours; Car 2 travels for $t_2 = t_1 - 1$ hours (starts 1 hr later, arrives same time).

Since both cover same distance D: $s_1 t_1 = s_2 t_2$.

Speed ratio: $s_1/s_2 = t_2/t_1$.

Percentage excess = $(s_2/s_1 - 1) \times 100 = (t_1/t_2 - 1) \times 100 = [1/t_2] \times 100$.

To maximise, minimise t_2 . Given $t_1 \geq 6$, minimum $t_1 = 6$, so minimum $t_2 = 5$.

Maximum percentage = $(1/5) \times 100 = 20\%$.

Question 30

A man leaves his home and walks at a speed of 12 km per hour, reaching the railway station 10 minutes after the train had departed. If instead he had walked at a speed of 15 km per hour, he would have reached the station 10 minutes before the train's departure. The distance (in km) from his home to the railway station is

TITA (Type in the answer)

Solution:

Let D = distance (km), T = scheduled departure time (hours from home).

At 12 km/hr: arrives 10 min late $\rightarrow D/12 = T + 1/6 \dots(1)$

At 15 km/hr: arrives 10 min early $\rightarrow D/15 = T - 1/6 \dots(2)$

Subtract (2) from (1):

$$D/12 - D/15 = 1/3$$

$$(5D - 4D)/60 = 1/3$$

$$D/60 = 1/3$$

$$D = 20 \text{ km.}$$

Question 31

Two ants A and B start from a point P on a circle at the same time, with A moving clock-wise and B moving anti-clockwise. They meet for the first time at 10:00 am when A has covered 60% of the track. If A returns to P at 10:12 am, then B returns to P at

A) 10:25 AM

B) 10:45 AM

C) 10:18 AM

✓ D) 10:27 AM ← Correct Answer

Solution:

At first meeting: A covered 60%, B covered 40% of track. Same time elapsed.

Speed ratio: $\text{Speed(A)}/\text{Speed(B)} = 60/40 = 3/2$.

A returns to P at 10:12 am. At first meeting (10:00 am), A had 40% of track left.

Time for A to cover 40% = 12 minutes.

$\text{Speed(A)} = 0.4 \times \text{Track} / 12 \text{ min.}$

$\text{Speed(B)} = (2/3) \times \text{Speed(A)} = (2/3) \times (0.4T/12) = 0.8T/36$.

At 10:00 am, B has covered 40%; B needs to cover remaining 60%.

Time for B = $0.6T / \text{Speed(B)} = 0.6T / (0.8T/36) = 0.6 \times 36/0.8 = 27 \text{ minutes.}$

B returns to P at 10:00 am + 27 min = 10:27 AM.

Question 32

Point P lies between points A and B such that the length of BP is thrice that of AP. Car 1 starts from A and moves towards B. Simultaneously, car 2 starts from B and moves towards A. Car 2 reaches P one hour after car 1 reaches P. If the speed of car 2 is half that of car 1, then the time, in minutes, taken by car 1 in reaching P from A is

TITA (Type in the answer)

Solution:

Let $AP = d$, $BP = 3d$. Speed of Car 1 = s ; Speed of Car 2 = $s/2$.

Time for Car 1 to reach P: $t_1 = d/s$.

Time for Car 2 to reach P: $t_2 = 3d/(s/2) = 6d/s$.

Car 2 arrives 1 hour (60 min) after Car 1:

$t_2 = t_1 + 60 \text{ min (in consistent units, 1 hour)}$

$6d/s = d/s + 1$

$5d/s = 1$

$d/s = 1/5 \text{ hour} = 12 \text{ minutes.}$

Time for Car 1 = 12 minutes.

Question 33

A man travels by a motor boat down a river to his office and back. With the speed of the river unchanged, if he doubles the speed of his motor boat, then his total travel time gets reduced by 75%. The ratio of the original speed of the motor boat to the speed of the river is

- A) $\sqrt{6} : \sqrt{2}$
✓ B) $\sqrt{7} : 2$ ← Correct Answer
 C) $2\sqrt{5} : 3$
 D) $3 : 2$

Solution:

Let u = original boat speed (still water), v = river speed, d = one-way distance.

Original total time $T_{\text{orig}} = d/(u+v) + d/(u-v) = 2du/(u^2-v^2)$.

New speed = $2u$. New total time $T_{\text{new}} = d/(2u+v) + d/(2u-v) = 4du/(4u^2-v^2)$.

$T_{\text{new}} = 0.25 \times T_{\text{orig}}$ (reduced by 75%):

$$4du/(4u^2-v^2) = (1/4) \times 2du/(u^2-v^2)$$

$$4/(4u^2-v^2) = 1/(2(u^2-v^2))$$

$$8(u^2-v^2) = 4u^2-v^2$$

$$8u^2-8v^2 = 4u^2-v^2$$

$$4u^2 = 7v^2$$

$$u^2/v^2 = 7/4$$

$$u/v = \sqrt{7/2}.$$

$$\text{Ratio} = \sqrt{7} : 2.$$

Question 34

In a race of three horses, the first beat the second by 11 metres and the third by 90 metres. If the second beat the third by 80 metres, what was the length, in metres, of the racecourse?

TITA (Type in the answer)

Solution:

Let L = length of racecourse (metres).

Speed ratios (all running simultaneously):

$$\bullet v_1/v_3 = (L-11)/(L-90) \dots(1) \text{ [from horse 1 vs 3 when horse 1 finishes]}$$

$$\bullet v_2/v_3 = L/(L-80) \dots(2) \text{ [from horse 2 vs 3]}$$

Equate (1) and (2):

$$(L-11)/(L-90) = L/(L-80)$$

$$(L-11)(L-80) = L(L-90)$$

$$L^2-80L-11L+880 = L^2-90L$$

$$-91L+880 = -90L$$

$$880 = L.$$

Length of racecourse = 880 metres.

Question 35

Train T leaves station X for station Y at 3 pm. Train S, traveling at three quarters of the speed of T, leaves Y for X at 4 pm. The two trains pass each other at a station Z, where the distance

between X and Z is three-fifths of that between X and Y. How many hours does train T take for its journey from X to Y?

TITA (Type in the answer)

Solution:

Let speed of T = v_T ; speed of S = $(3/4)v_T$. Total distance X to Y = D.

Let trains meet t hours after 3 pm.

Train T travels for t hours: distance = $v_T \times t = (3/5)D$...(1)

Train S travels for (t-1) hours: distance = $(3/4)v_T \times (t-1) = (2/5)D$...(2)

Divide (1) by (2):

$$t / [(3/4)(t-1)] = (3/5)/(2/5) = 3/2$$

$$4t / [3(t-1)] = 3/2$$

$$8t = 9(t-1)$$

$$8t = 9t - 9$$

$$t = 9 \text{ hours.}$$

Total time for T to cover D:

$$\text{From (1): } v_T \times 9 = (3/5)D \rightarrow D = 15v_T \times 9/9 \dots$$

$$T_{\text{total}} = D/v_T = (3/5)D / [(3/5)v_T] \dots$$

$$(3/5)T_{\text{total}} = 9 \rightarrow T_{\text{total}} = 15 \text{ hours.}$$

Question 36

One can use three different transports which move at 10, 20, and 30 kmph, respectively to reach from A to B. Amal took each mode of transport for 1/3rd of his total journey time, while Bimal took each mode of transport for 1/3rd of the total distance. The percentage by which Bimal's travel time exceeds Amal's travel time is nearest to

✓ A) 22 ← Correct Answer

B) 20

C) 19

D) 21

Solution:

Amal (equal time):

$$\text{Total distance } D = (T_A/3)(10 + 20 + 30) = (T_A/3)(60) = 20 \cdot T_A.$$

Bimal (equal distance):

$$T_B = (D/3)(1/10 + 1/20 + 1/30) = (D/3)(6+3+2)/60 = (D/3)(11/60) = 11D/180.$$

Substitute $D = 20T_A$:

$$T_B = 11(20T_A)/180 = 220T_A/180 = 11T_A/9.$$

$$\% \text{ excess} = (T_B - T_A)/T_A \times 100 = (11/9 - 1) \times 100 = (2/9) \times 100 \approx 22.2\% \approx 22.$$

Question 37

A cyclist leaves A at 10 am and reaches B at 11 am. Starting from 10:01 am, every minute a motorcycle leaves A and moves towards B. Forty-five such motorcycles reach B by 11 am. All motorcycles have the same speed. If the cyclist had doubled his speed, how many motorcycles

would have reached B by the time the cyclist reached B?

- A) 22
- B) 23
- ✓ C) 15 ← Correct Answer
- D) 20

Solution:

Original cyclist: leaves 10:00, arrives 11:00 (60 min journey).
45th motorcycle leaves at $10:01 + 44 = 10:45$ am, arrives by 11:00 am.
Travel time of each motorcycle = $11:00 - 10:45 = 15$ minutes.
Doubled speed: cyclist arrives at $10:00 + 30 \text{ min} = 10:30$ am.
A motorcycle starting at time $(10:00 + t)$ arrives at $(10:00 + t + 15 \text{ min})$.
For arrival $\leq 10:30$ am: $t + 15 \leq 30 \rightarrow t \leq 15$ minutes.
Motorcycles that left at 10:01, 10:02, ..., 10:15 $\rightarrow 15$ motorcycles.

Question 38

John jogs on track A at 6 kmph and Mary jogs on track B at 7.5 kmph. The total length of tracks A and B is 325 metres. While John makes 9 rounds of track A, Mary makes 5 rounds of track B. In how many seconds will Mary make one round of track A?

TITA (Type in the answer)

Solution:

John and Mary run for the same time T .
 $9 \times L_A / 6 = 5 \times L_B / 7.5$
 $1.5 L_A = (2/3) L_B \rightarrow L_A / L_B = 4/9$.
 $L_A + L_B = 325 \text{ m}$; ratio 4:9 (total 13 parts).
 $L_A = (4/13) \times 325 = 100 \text{ m}$.
Time for Mary to run $L_A = 100 \text{ m}$ at 7.5 kmph:
 $= 0.1 \text{ km} / 7.5 \text{ kmph} = 1/75 \text{ hours} = 3600/75 = 48 \text{ seconds}$.

Question 39

The distance from A to B is 60 km. Partha and Narayan start from A at the same time and move towards B. Partha takes four hours more than Narayan to reach B. Moreover, Partha reaches the mid-point of A and B two hours before Narayan reaches B. The speed of Partha, in km per hour, is

- A) 6
- B) 4
- C) 3
- ✓ D) 5 ← Correct Answer

Solution:

Let Narayan's time = t_n hours; Partha's time = $t_n + 4$ hours.
Speeds: $s_n = 60/t_n$; $s_p = 60/(t_n + 4)$.
Partha reaches midpoint (30 km) two hours before Narayan reaches B:

$$30/s_p = t_n - 2$$

$$30 / [60/(t_n+4)] = t_n - 2$$

$$(t_n+4)/2 = t_n - 2$$

$$t_n + 4 = 2t_n - 4$$

$$t_n = 8 \text{ hours.}$$

$$\text{Partha's time} = 12 \text{ hours. Speed} = 60/12 = 5 \text{ km/hr.}$$

Question 40

Ravi is driving at a speed of 40 km/h on a road. Vijay is 54 meters behind Ravi and driving in the same direction as Ravi. Ashok is driving along the same road from the opposite direction at a speed of 50 km/h and is 225 meters away from Ravi. The speed, in km/h, at which Vijay should drive so that all three cross each other at the same time, is

A) 58.8

B) 67.2

✓ C) 61.6 ← Correct Answer

D) 64.4

Solution:

Convert to m/s: Ravi = 100/9 m/s; Ashok = 125/9 m/s.

Ravi and Ashok are 225 m apart, moving toward each other.

Relative speed = $100/9 + 125/9 = 225/9 = 25 \text{ m/s}$.

Time to meet = $225/25 = 9 \text{ seconds}$.

In 9 seconds Ravi travels = $(100/9) \times 9 = 100 \text{ m}$ from his initial position.

Meeting point is 100 m ahead of Ravi's start.

Vijay is 54 m behind Ravi's start, so must cover $54 + 100 = 154 \text{ m}$ in 9 seconds.

Vijay's speed = $154/9 \text{ m/s} = (154/9) \times (18/5) \text{ km/h} = 154 \times 2/5 = 308/5 = 61.6 \text{ km/h}$.

Question 41

A train travelled a certain distance at a uniform speed. Had the speed been 6 km per hour more, it would have needed 4 hours less. Had the speed been 6 km per hour less, it would have needed 6 hours more. The distance, in km, travelled by the train is

✓ A) 720 ← Correct Answer

B) 800

C) 780

D) 640

Solution:

Let distance = D, speed = S km/hr, time = T hours. $D = S \times T$.

Condition 1: $(S+6)(T-4) = D$

$ST - 4S + 6T - 24 = ST$

$-4S + 6T = 24 \rightarrow 3T - 2S = 12 \dots(1)$

Condition 2: $(S-6)(T+6) = D$

$ST + 6S - 6T - 36 = ST$

$6S - 6T = 36 \rightarrow S - T = 6 \rightarrow S = T + 6 \dots(2)$

Substitute (2) into (1):

$$3T - 2(T+6) = 12$$

$$3T - 2T - 12 = 12$$

$$T = 24 \text{ hours.}$$

$$S = 24 + 6 = 30 \text{ km/hr.}$$

$$D = 30 \times 24 = 720 \text{ km.}$$
